

Prognostic Value of HER-2/neu Gene Amplification in Epithelial Ovarian Carcinoma

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Abstract

Background Human epidermal growth factor receptor 2 (HER2) has tyrosine kinase activity and is a member of the epidermal growth factor receptor family. This initiates a variety of signal for pathways which leads to cell proliferation and tumorigenesis. In approximately 15–30% of breast cancers and 10–30% of gastric/gastroesophageal cancers, amplification or overexpression of HER2 occurs, and it serves as a prognostic and predictive biomarker. HER-2/neu is one of the most frequently studied molecular biological parameters in epithelial ovarian cancer (EOC), but its prognostic impact has not been fully assessed. The objective of the current study was firstly to analyse the presence of HER-2 overexpression in EOC patients and secondly to evaluate association between human epidermal growth factor receptor 2 (HER-2/neu) expression and progression-free survival in patients with EOC.

Method In this prospective study of 2 years in our department, 186 gynaecological cancers were operated out of which 98 cases were operated for epithelial ovarian cancer and rest for cervical cancer, endometrial cancer, and other subtypes of ovarian cancer. In this study, *HER2* gene status was evaluated in EOC by immunohistochemistry analysis, and overall survival of these patients was evaluated.

Results HER-2 overexpression was found in 22 of the 98 investigated cases (22.45%), in which 14 OC (14.29%) were weakly positive and 8 OC (8.16%) were moderately to intensely positive. In the study, increased HER2 expression was associated with decreased progression-free survival (PFS) and hence poor prognosis (P 0.054).

Conclusions The introduction of HER2-directed therapies has dramatically influenced the outcome of patients with HER2 positive breast and gastric/gastroesophageal cancers. The present study findings provided that HER-2/neu expression in patients with OC has an adverse impact on the PFS. Therefore, our results show that the decision algorithm usually used in breast cancer by HER2 may be appropriate in ovarian cancer.

Keywords

Human epidermal growth factor receptor 2 (HER-2) · Immunohistochemistry (IHC) · Ovarian carcinoma · Prognosis · Overall survival · Progression-free survival

Introduction

Epithelial ovarian cancer is the fourth most common cancer after breast, cervix, and colorectal cancer in India. In GLOBOCON 2012, age-standardised incidence rate of ovarian cancer is 4.9 and age standard mortality rate is 3.6. It is the most frequently encountered cause of gynaecological cancer death and the fifth leading cause of cancer deaths in developing countries [1]. A total of 21,880 new cases of ovarian cancer were diagnosed in the US which resulted in 13,850 deaths during 2010, with a cure rate of less than 40% [2]. The relatively poor prognosis of ovarian cancer is due to lack of detection at an early stage and limited application of effective therapies for the advanced-stage disease [3]. Established clinical-pathological prognostic factors of EOC include World Health Organization (WHO) grade, residual tumour after primary surgery, age at diagnosis, performance status, histological characteristics, and tumour rupture during surgery. However, these factors inadequately predicted the clinical outcomes of EOC. Recently, several molecular markers that contribute significant roles in the formation and progression of EOC have been identified. According to the NCCN Ovarian Cancer Guideline Version 1.2016, BRCA1 or BRCA2

mutations are high risk carriers of ovarian cancer [4]. A UK-based trial compared the multimodality screening of ovarian cancer with ultrasound and CA-125 versus either ultrasound alone and/or no screening and concluded that the former was more effective in the detection of early-stage cancer [5]. Despite this evidence, the molecular mechanisms contributing to its aggressiveness are not fully understood. Therefore, the identification of novel prognostic markers has a substantial clinical impact on the future management of EOC. P53 is a tumour suppressor protein that is widely studied as a prognostic factor of EOC. In addition, the EGFR and HER-2/neu proteins are considered prognostic factors of EOC. These markers have been used in cancer therapy. HER-2 (p185, HER-2/neu, ErbB-2) is a tyrosine kinase receptor that is located in the cell membrane and is 185 kDa in size. HER-2 receptor regulates a multitude of biological processes such as cell proliferation, apoptosis, survival, migration, and differentiation [6]. The majority of the subsequent studies further support the hypothesis that HER-2/neu overexpression has an adverse prognostic factor for the survival of patients with EOC.

Therefore, this study was conducted to explore the effects of HER-2/neu expression on the progression-free survival in EOC.

Materials and Methods

This prospective study was done in 98 patients of EOC who had undergone debulking and staging laparotomy in IGIMS from September 2015 to August 2017. Out of 98 patients of EOC, 48 had received neoadjuvant chemotherapy with carboplatin and paclitaxel before surgery and 50 patients of EOC had undergone upfront surgery. HER2 status was done by IHC of paraffin-embedded tissue on conventional slides. All the patients were followed up during this period, and progression-free survival was evaluated.

Among the numerous methods published, immunohistochemistry (IHC) and fluorescence in situ hybridization (FISH) are most widely used and have highly reported concordance [7]. FISH has been shown to be more accurate in selected patients than IHC but is more costly and not routinely available. The recommended algorithm for *HER2* determination in breast cancer is to use IHC initially, using a semi-quantitative scoring system followed by FISH for 2 + ambiguous samples [8].

HER2 positivity was assessed using Ellis and Wolff recommendations. A score of 1 + was defined as barely perceptible membrane staining in more than 10% of cells, a score of 2 + was defined as weak-to-moderate complete membrane with staining present in more than 10% of

tumour cells, and a score of 3 + was defined as strong complete membrane staining in more than 10% of tumour cells. We classified 2 + as equivocal and 3 + as positive. Cytoplasmic staining was considered to be non-specific.

Result

Of total of 98 EOC cases, the expression pattern for 76 cases (77.55%) was negative (score of 0), 11(11.22%) patients were weakly positive (score of 1 +) for 15 cases, and 11 (11.22%) patients were moderately to intensely positive (score of 2 +, score of 3 +). For each stage of OC, the distribution of cases in relation to the expression level (0, 1 +, and 2 +/3 +) is stated in Table 1. Overall, we found that HER-2 was expressed in the OC tissue of 22 tumours (22.45%), scored as either 1 +, 2 +, or 3 +. We observed an increased tendency towards HER-2 expression with increasing tumour grade, but no significant correlation was found between HER-2 expression and staging of OC (P 0.58). Expression of HER-2 was examined in tissue from patients with invasive EOC of different histology (Table 1). We did not find any correlation between HER-2 expression and histological type of tumour (papillary adenocarcinoma, serous adenocarcinoma, mucinous adenocarcinoma, clear cell carcinoma, and endometrioid adenocarcinoma) (P 0.55). The median age of the patients was 50 years (range, 35–70 years). No significant difference was found with regard to the median age of OC patients with positive (1 +, 2 +, or 3 +) and negative HER-2 expression in the tumour tissue (P 0.12). Univariate survival analysis was performed on FIGO Stage I, Stage II, Stage III, and Stage IV OC subgroups. The relation

between HER-2 OC tissue expression and survival was analysed in 12 patients with Stage I disease, 34 patients with Stage II disease, 33 patients with Stage III disease, and 19 patients with Stage IV disease. Significant differences in survival were found between patients who were HER-2-positive (score of 1, 2, or 3) and those who were HER-2-negative (12 Stage I OC patients [P 0.021], 34 Stage II OC patients [P 0.078], 33 Stage III OC patients [P 0.0064], and 19 Stage IV OC patients [P 0.0054] Table 2. No significant difference in survival was found in Stage I and Stage II OC Patients. Histological type of tumour (P 0.47) and age (P 0.63) were found to have no independent prognostic value. Recurrence and mortality were highest with Stage III and IV OC. PFS was highest in Stage I patients. Follow-up of all patients was done; 12 patients did not turn up after 18 months, 08 died within 12 months, 24 had recurrent disease after 24 months, and 54 patients were alive and on follow-up at 24 months.

Discussion

In this study, we evaluated the effects of HER2 expression together with clinical-pathological factors on the prognosis of EOC patient's population in which the majority of the patients had advanced-stage disease. We found 22.45% patients were HER2-positive which was compatible with the results of previous studies with in EOC as a prognostic factor [7, 8].

Among the numerous methods published, immunohistochemistry (IHC) and fluorescence in situ hybridization (FISH) are most widely used and have high reported concordance. FISH has been shown to be more accurate than

Table 1 Immunohistochemical staining for HER-2 in relation to clinical tumour stage and histological characteristics in the current study

Epithelial ovarian carcinoma	Total (<i>n</i> = 98)	Immunostaining pattern of the tumour tissue		
		HER2 (–)ve (<i>n</i> = 76)	HER2 (scoring 1 +, 2 +/3 +) (<i>n</i> = 22)	
			1 +	2 +/3 +
<i>OC</i>				
Stage I	12 (12.24%)	10 (83.33%)	1 (8.33%)	1 (8.33%)
Stage II	34 (34.69%)	28 (82.35%)	3 (8.82%)	3 (8.82%)
Stage III	33 (33.67%)	26 (78.79%)	3 (9.09%)	4 (12.12%)
Stage IV	19 (19.39%)	12 (63.16%)	4 (21.05%)	3 (15.78%)
<i>Tumour cell type</i>				
Poorly differentiated carcinoma	2 (2.04%)	1 (50%)	1 (50%)	
Papillary adenocarcinoma	19 (19.39%)	14 (73.68%)	5 (26.31%)	
Serous adenocarcinoma	58 (59.18%)	48 (82.76%)	10 (17.24%)	
Mucinous adenocarcinoma	17 (17.35%)	12 (70.59%)	5 (29.41%)	
Endometrioid adenocarcinoma	1 (1.02%)	1 (100%)	–	
Clear cell neoplasm	1 (1.02%)	0	1 (100%)	

Table 2 Distribution of patients according to treatment given and survival analysis for 2 years

EOC (<i>n</i> = 98)	Upfront surgery (<i>n</i> = 50)	Surgery after neoadjuvant (<i>n</i> = 48)	HER2 status	Recurrence (<i>n</i> = 13)	Progression-free survival (PFS) (<i>n</i> = 69)	Mortality (<i>n</i> = 12)	Loss to follow- up (<i>n</i> = 4)
Stage I (<i>n</i> = 12)	11 (91.67%)	1 (8.33%)	HER2 (–)ve (<i>n</i> = 10)	–	2	–	–
			HER2 (+)ve (<i>n</i> = 2)	–	10	–	–
Stage II (<i>n</i> = 34)	25 (73.53%)	9 (26.47%)	HER2 (–)ve (<i>n</i> = 28)	2	24	1	1
			HER2 (+)ve (<i>n</i> = 6)	1	4	1	–
Stage III (<i>n</i> = 33)	12 (36.36%)	21 (63.63%)	HER2 (–)ve (<i>n</i> = 26)	4	19	1	2
			HER2 (+)ve (<i>n</i> = 7)	2	2	3	–
Stage IV (<i>n</i> = 19)	2 (10.53%)	17 (8.47%)	HER2 (–)ve (<i>n</i> = 12)	3	6	2	1
			HER2 (+)ve (<i>n</i> = 7)	1	2	4	–

IHC but is more costly and not routinely available. In the present study immunohistochemical (IHC) approach with a polyclonal antibody was used to determine the expression of HER-2 in paraffin-embedded tissues from epithelial ovarian carcinomas. IHC is a reliable diagnostic technique due to its high sensitivity and specificity, its simplicity and cost-effectiveness.

By using Ellis and Wolff recommendations [7] *HER2* positivity was assessed as scoring. A score of 1 + was defined as barely perceptible membrane staining in more than 10% of cells, a score of 2 + was defined as weak-to-moderate complete membrane with staining present in more than 10% of tumour cells, and a score of 3 + was defined as strong complete membrane staining in more than 10% of tumour cells. Cytoplasmic staining was considered to be non-specific. We classified 1 + and 2 + as equivocal and 3 + as positive. For optimal estimation of the staining intensity, two experienced persons should evaluate the staining results independently. However, the results obtained by IHC are highly dependent on the storage time and fixation method of the paraffin-embedded tissues, the selection of the primary antibody and the IHC staining protocol [9, 10].

As compare to 20–30% rate of *HER2* positivity observed in breast cancer, the rate in ovarian cancer is lower and intratumoural heterogeneity is frequently detected. *HER2* positivity varies in the literature from 8 to 66%. This wide variation is due to different detection methods used (IHC, FISH, and chromogenic in situ hybridization), different sources of material (blocks of tumours and tissue microarray), and variations in IHC assay techniques (CB-11, HercepTest or a non-commercial

antibody); in addition, variance in staining protocols and subjective interpretation of sample stains makes direct comparison of studies difficult. In this study for IHC staining, tissue blocks were sent to a single lab where a single IHC assay techniques were used, and the result was interpreted by two experienced persons.

HER2 is a member of the receptor tyrosine kinase family and plays a crucial role in the growth of both normal tissue and malignant tumours [6]. *HER2* overexpression is observed in 18.9–22.4% of epithelial ovarian carcinomas, and it is reported to be seen more frequently in mucinous ovarian cancers [11]. The prognostic role of *HER2* overexpression in EOC remains unclear. Some studies proposed that an elevated *HER2* level was associated with poorer survival, whereas other studies could not demonstrate any association. In the meta-analysis conducted by Zhao et al. [12], the prognostic effect of *HER2* overexpression was also explored; after reviewing eight studies, they demonstrated that *HER2* overexpression was significantly associated with poorer overall survival. In our study, *HER2* positivity was observed in 22.45% of the cases, and the majority of *HER2*-positive tumours were serous tumours (10/22), while only 5/22 cases had mucinous histology. *HER2* overexpression was significantly associated with shorter progression-free survival specifically in patients with advanced-stage disease [Stage III (*P* 0.0064) Stage IV (*P* 0.0054)]. Additionally, *HER2*-positive patients tended to have shorter PFS times, suggesting its negative predictive effect. Cancer phenotypic and molecular subtypes have been widely used in predicting the treatment response and survival of breast cancer patients in recent decades. However, after the development of trastuzumab, a monoclonal

antibody against the HER2 receptor, the prognosis of HER2-positive breast cancer has been improved. Due to the lack of any targeted therapy, triple-negative breast cancer (ER, PR, HER2) still remains the most aggressive subtype in breast cancer. In contrast, anti-HER2 therapy in HER2(+) EOC is not as effective as in breast cancer. Two previous phase II studies failed to show benefit from trastuzumab in HER2-positive EOC [13–15]. Moreover, trastuzumab has not been widely used for ovarian carcinoma due to the limited number of studies in this area.

To summarise, this study pointed out the following findings: HER2 positivity significantly shortens progression-free survival in patients with advanced-stage disease, and the HER2 positive phenotype has more aggressive behaviour.

Conclusions

Taken collectively, the results of the present study provide strong evidence of the prognostic value of HER-2/neu expression for EOC patients. The study further indicated that HER-2/neu is associated with lower PFS. Therefore, it can be deduced that HER-2/neu-targeted therapy is a promising and revolutionary strategy for cancer patients.

Compliance with Ethical Standards

Conflict of interest We declare that this manuscript is original; it has not been published anywhere before and is not currently being considered for publication elsewhere. We also confirm that the authors do not have any conflict of interest associated with publication of this work and no significant financial support/funding for this work has been received to influence the outcome. The manuscript is read, approved, and consent is given by all the authors. We give our permission to reproduce any material of the article.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent was obtained from all individual participants included in the study. This article does not contain any studies with animals performed by any of the authors.

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